	AIR UNIVERSITY
	DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
	EXPERIMENT NO 1

Lab Title Introduction to Matlab

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Objective:

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks: _____

Date: _____ Signature: _____



Department of Electrical and Computer Engineering

Signal and Systems Lab

Lab Report: 1

Title: Introduction to

Matlab

Name: Ali Abbas Khan

Roll Number: 230661

Class: BCE-IV-B

Submitted To: Engr. Maryam

Date: September 19, 2025

Lab Tasks

1. Write the codes for the following tasks. Paste the codes and their results in the space provided.

1. Create vectors:

a. 'A' of even whole numbers between 19 and 63.

b. 'B' of odd whole numbers between 66 and 120.

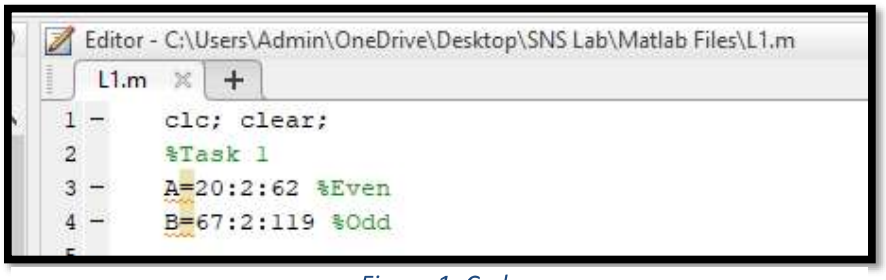


Figure 1: Code

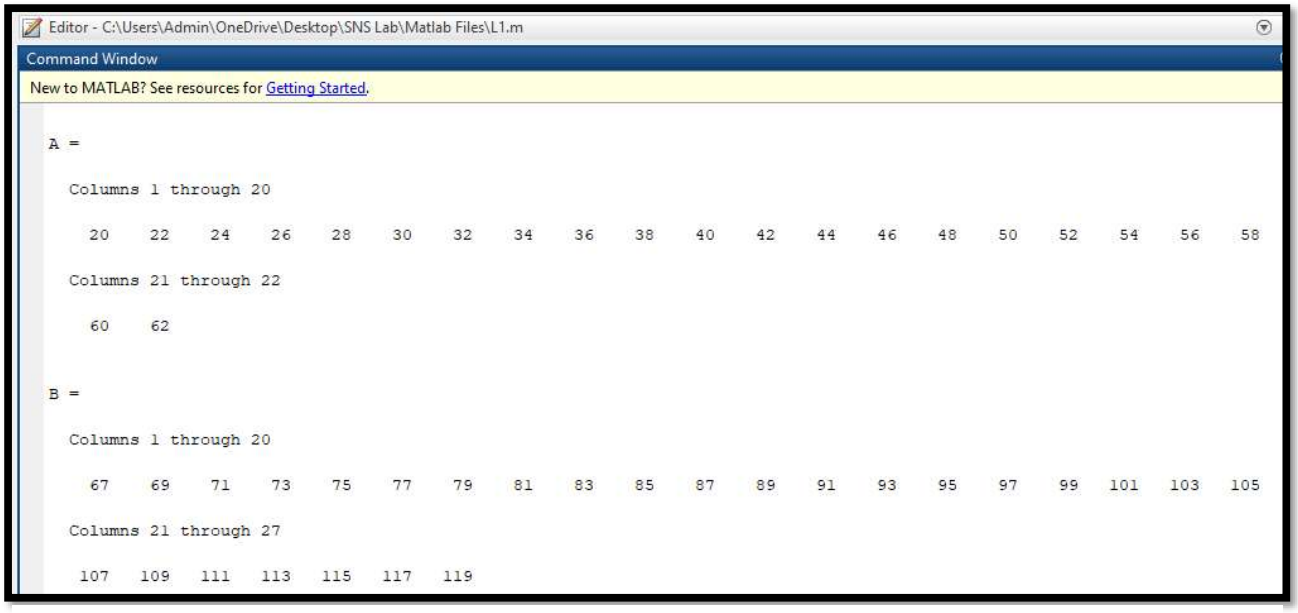


Figure 2: Output

2. Let x = [3 -2 0 5 1 9 5 6]

a. Add 5 to each element of this vector.
b. Multiply the even indexed elements with 4.
c. Taking square of each element.
d. Compute the cube of each element.

```
6       %Task 2
7 -     x=[3 -2 0 5 1 9 5 6]
8 -     y=x+5;   %Part a
9 -     c=x(2:2:8)*4 %Part b
10 -    d=x.^2   %Part c
11 -    e=x.^3   %Part d
12
```

Figure 3: Code

```
x =

     3    -2     0     5     1     9     5     6


y =

     8     3     5    10     6    14    10    11


c =

    -8    20    36    24


d =

     9     4     0    25     1    81    25    36


e =

    27    -8     0   125     1   729   125   216
```

Figure 4: Output

3. Let x = [4 3 9 1] and y = [2 1 9 -3]

a. Add the x and y elements.

b. Raise each element of y to the power specified by the corresponding element of x.

c. Multiply each element in x by the corresponding element in y and store the result in ‘c’.

d. Evaluate y*x’ and interpret the result.

```
%Task 3
x=[4 3 9 1]
y=[2 1 9 -3]
a=x+y %Part a
b=y.^x %Part b
c=x.*y %Part c
d=y*x' %Part d This Performs the dot product of the two vectors
```

Figure 5: Code

```
x =

     4     3     9     1


y =

     2     1     9    -3


a =

     6     4    18    -2


b =

          16           1   387420489          -3


c =

     8     3    81    -3


d =

    89
```

Figure 6: Output

4. Use randi command in MATLAB and generate a 5x5 matrix.

Use MATLAB commands to find sum of all elements, maximum number, minimum number, inverse and transpose of this matrix.

```
21      %Task4
22 -    A=randi(100,5,5) %Random Matrix
23 -    G=sum(A(:))  %Sum of All Elements
24 -    M=max(A(:))  %Maximum Number
25 -    m=min(A(:))  %Minimum Number
26 -    I=inv(A)     %Inverse of Matrix
27 -    T=A'         %Transpose of Matrix
```

Figure 7: Code

```
A =

    76    71    83    44    49
    75     4    70    39    45
    40    28    32    77    65
    66     5    96    80    71
    18    10     4    19    76


G =

        1243


M =

    96


m =

     4


I =

   -0.0044    0.0346    0.0125   -0.0254   -0.0046
    0.0140   -0.0116    0.0026   -0.0046   -0.0002
    0.0071   -0.0176   -0.0200    0.0238    0.0008
   -0.0060   -0.0020    0.0200    0.0006   -0.0125
    0.0003   -0.0052   -0.0072    0.0052    0.0174
```

Figure 8: Output First Half

```
T =

    76    75    40    66    18
    71     4    28     5    10
    83    70    32    96     4
    44    39    77    80    19
    49    45    65    71    76
```

Figure 9: Output Second Half

5. Given a matrix A, such that:

$$A = \begin{bmatrix} 5 & 8 & 9 & 6 \\ 8 & 5 & 8 & 6 \\ 4 & 9 & 5 & 8 \end{bmatrix}$$

Perform the following operations:
a. Extract the 3rd column of matrix A and store it in vector B.
b. Extract the 1st and 3rd columns of matrix A and store them in matrix C.
c. Add the 1st and 3rd rows of matrix A together and store the result in vector D.
d. Change the value in the 2nd row and 3rd column of A to 7 (instead of 5) and call the result
AA (do not destroy/change the original A matrix).
e. Create a matrix that contains rows 1 and 3 from A, the second row of AA, and the result of
part
(c).

```
29      %Task 5
30 -    A=[5 8 9 6;8 5 8 6;4 9 5 8]
31 -    B=A(:,3)  %Part a
32 -    C=[A(:,1),A(:,3)] %Part b
33 -    D=A(1,:)+A(3,:) %Part c
34 -    AA=A;
35 -    AA(2,3)=7  % Part d
36 -    H=[A(1,:);A(3,:);AA(2,:);D] %Part e
37
```

Figure 10: Code

```
A =

     5     8     9     6
     8     5     8     6
     4     9     5     8


B =

     9
     8
     5

C =

     5     9
     8     8
     4     5


D =

     9    17    14    14


AA =

     5     8     9     6
     8     5     7     6
     4     9     5     8
```

Figure 11: Output First Half

```
H =

     5     8     9     6
     4     9     5     8
     8     5     7     6
     9    17    14    14
```

Figure 12: Output Second Half

Conclusion:

In this lab, we explored MATLAB's fundamentals, focusing on **vector and matrix operations**. The exercises covered creating vectors with specific ranges, performing element-wise arithmetic, and using built-in functions to analyze data. We learned to manipulate matrices by finding sums, maximums, and minimums, as well as calculating

the inverse and transpose. The lab also demonstrated how to extract and modify specific parts of a matrix using indexing. This practical experience highlighted MATLAB's efficiency as a tool for numerical computation.